

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)	Substitution: $W = \frac{1}{2} \times 1 \times 0.4$ or $W = \frac{1}{2} \times 2.5 \times 0.4^2$ (1) $W = 0.2$ [J] (1)	1	1		2	2	
		(ii)	Either: $W = \frac{1}{2}Fx$ (1) F doubles and x doubles or W increased $\times 4$ or increased to 0.8 J (1) Or: $W = \frac{1}{2}kx^2$ (1) k constant (by implication) and x is squared or W increased $\times 4$ or increased to 0.8 J (1) Or: Correct use of numerical values from graph e.g. Area under graph stated or implied (1) W increased $\times 4$ or increased to 0.8 J or equivalent (1)	1	1		2	1	